

SHETH L.U.J & SIR MV COLLEGE

Practical 2: RSA Encryption and Decryption

Aim: Implement the RSA algorithm for public-key encryption and decryption, and explore

its properties and security considerations.

NOTE: Use both the GUI and CLI methods.

Code:CLI

```
from math import gcd

def mod_inverse(e, phi):
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d

p = 61
q = 53

n = p * q
phi = (p - 1) * (q - 1)

e = 17
while gcd(e, phi) != 1:
    e += 2

d = mod_inverse(e, phi)

def encrypt(message):
    return [pow(ord(ch), e, n) for ch in message]

def decrypt(cipher):
    return ''.join(chr(pow(i, d, n)) for i in cipher)

while True:
    print("\n----- RSA MENU -----")
    print("1. Encrypt")
    print("2. Decrypt")
    print("3. Exit")
```

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

```
choice = input("Enter choice: ")
```

```
if choice == "1":
```

```
    message = input("Enter message: ")
```

```
    cipher = encrypt(message)
```

```
    print("Encrypted Data:")
```

```
    print(*cipher)
```

```
elif choice == "2":
```

```
    data = input("Enter encrypted numbers (space separated): ")
```

```
    try:
```

```
        cipher = list(map(int, data.split()))
```

```
        print("Decrypted Message:", decrypt(cipher))
```

```
    except:
```

```
        print("Invalid input.")
```

```
elif choice == "3":
```

```
    print("Program Closed.")
```

```
    break
```

```
else:
```

```
    print("Invalid Choice.")
```

SHETH L.U.J & SIR MV COLLEGE

Output:

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/IT-23/Desktop/purvi T086/cyber security/prac2-T086-cyber security-cli.py

----- RSA MENU -----
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 1
Enter message: PURVI KARKERA
Encrypted Data:
2933 2310 1859 1906 1486 1992 597 2790 1859 597 28 1859 2790

----- RSA MENU -----
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 2
Enter encrypted numbers (space separated): 2933 2310 1859 1906 1486 1992 597 2790 1859 597 28 1859 2790
Decrypted Message: PURVI KARKERA

----- RSA MENU -----
1. Encrypt
2. Decrypt
3. Exit
Enter choice: 3
Program Closed.
>>> = RESTART: C:/Users/IT-23/Desktop/purvi T086/cyber security/prac2-T086-cyber security-GUI.py
```

Code:GUI

```
import tkinter as tk
from tkinter import messagebox
from math import gcd
```

```
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
```

```
p = 61
q = 53
```

```
n = p * q
phi = (p - 1) * (q - 1)
```

```
e = 17
while gcd(e, phi) != 1:
    e += 2
```

Purvi.R.Karkera
T086

SHETH L.U.J & SIR MV COLLEGE

```
d = mod_inverse(e, phi)
```

```
def encrypt():
```

```
    message = input_box.get()
```

```
    if message == "":
```

```
        messagebox.showwarning("Warning", "Please enter a message.")
```

```
    return
```

```
    cipher = [str(pow(ord(ch), e, n)) for ch in message]
```

```
    output_box.delete("1.0", tk.END)
```

```
    output_box.insert(tk.END, " ".join(cipher))
```

```
def decrypt():
```

```
    try:
```

```
        cipher = list(map(int, input_box.get().split()))
```

```
        message = "".join(chr(pow(num, d, n)) for num in cipher)
```

```
        output_box.delete("1.0", tk.END)
```

```
        output_box.insert(tk.END, message)
```

```
    except:
```

```
        messagebox.showerror("Error", "Invalid encrypted data.")
```

```
def clear():
```

```
    input_box.delete(0, tk.END)
```

```
    output_box.delete("1.0", tk.END)
```

```
root = tk.Tk()
```

```
root.title("RSA Encryption and Decryption")
```

```
root.geometry("500x320")
```

```
root.resizable(False, False)
```

```
root.configure(bg="#F5F5F5")
```

```
title = tk.Label(
```

```
    root,
```

```
    text="RSA Encryption and Decryption",
```

```
    font=("Arial", 16, "bold"),
```

```
    bg="#F5F5F5"
```

```
)
```

```
title.pack(pady=15)
```

```
label = tk.Label(
```

```
    root,
```

Purvi.R.Karkera

T086

SHETH L.U.J & SIR MV COLLEGE

```
text="Enter Message or Encrypted Numbers",
font=("Arial", 11),
bg="#F5F5F5"
)
label.pack()

input_box = tk.Entry(root, width=55, font=("Arial", 11))
input_box.pack(pady=10)

button_frame = tk.Frame(root, bg="#F5F5F5")
button_frame.pack()

tk.Button(
    button_frame,
    text="Encrypt",
    width=12,
    command=encrypt
).grid(row=0, column=0, padx=8)

tk.Button(
    button_frame,
    text="Decrypt",
    width=12,
    command=decrypt
).grid(row=0, column=1, padx=8)

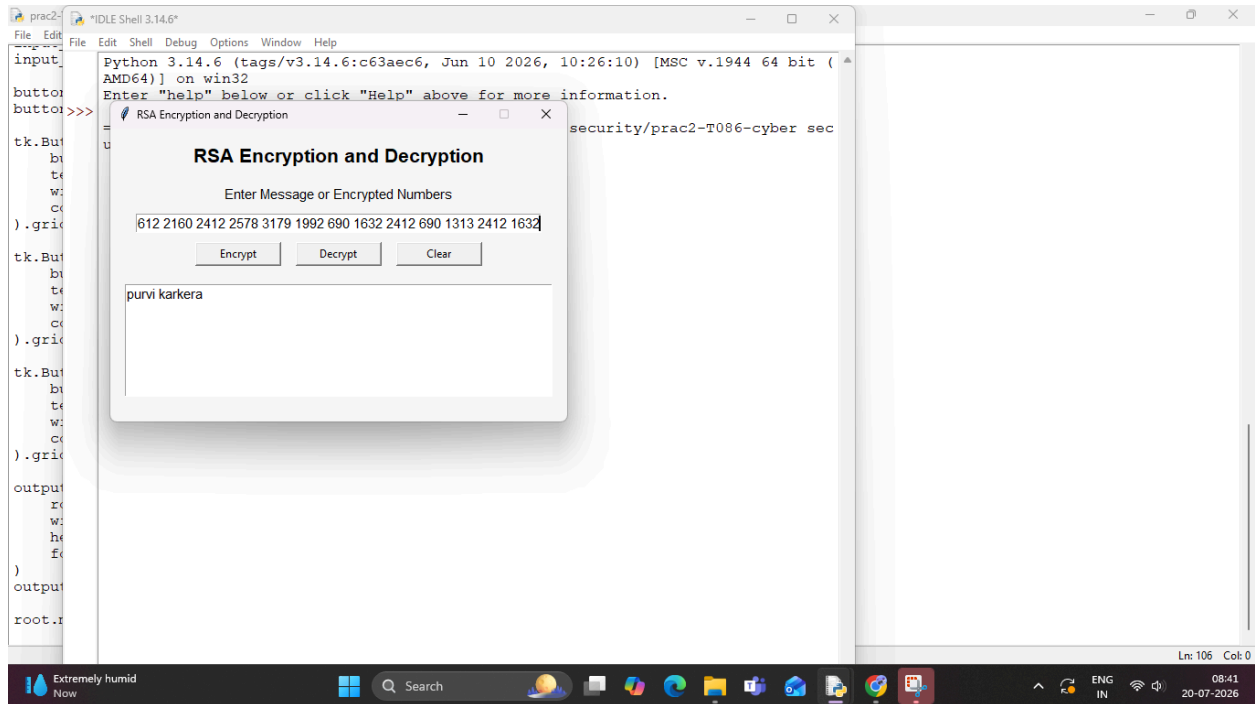
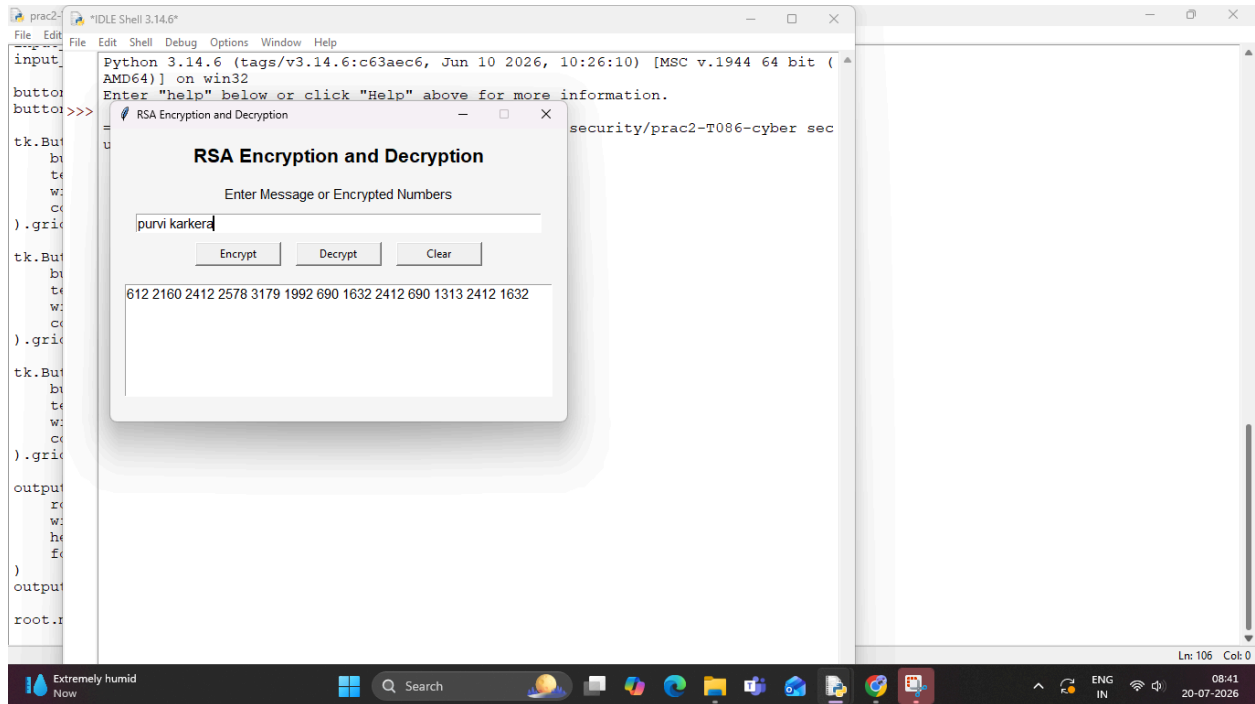
tk.Button(
    button_frame,
    text="Clear",
    width=12,
    command=clear
).grid(row=0, column=2, padx=8)

output_box = tk.Text(
    root,
    width=58,
    height=7,
    font=("Arial", 11)
)
output_box.pack(pady=20)

root.mainloop()
```

SHETH L.U.J & SIR MV COLLEGE

Output:



Purvi.R.Karkera
T086